

Contractive Projection Dynamics on Hilbert Spaces

Foundation for the Rigid Identity Framework

Johnny Andrey Pérez Ramírez

Abstract

We establish a canonical mathematical foundation for contractive dynamical systems on Hilbert spaces with metric projections onto closed convex subsets. We prove existence and uniqueness of projections, contractivity of the induced transition operator, existence of unique attractors, global compactness, and structural non-collapsibility. All results are derived from four minimal hypotheses (H1–H4) using standard functional analysis. This core serves as the rigorous foundation for the Rigid Identity Framework, where *Rigid Identity* and *Phenomenological Regimes* become direct corollaries and the mathematical dependencies used by *Null-Regime Instability* are made explicit. No ontological, phenomenological, or semantic interpretations are assumed; all results are purely structural.

Release governance note. Methods, governance conventions, claim boundaries, and terminology policy for this release are maintained in the project’s canonical governance hub (*Methods Governance Hub*, release version 1). This document adheres to that hub as its editorial and methodological reference.

Contents

1	Introduction	3
1.1	Purpose	3
1.2	Methodology	3
1.3	Scope	3
2	Minimal Hypotheses	3
2.1	Necessity and Sufficiency	3
3	The Projection Operator	4
4	The Transition Operator	5
5	Existence of Unique Attractors	5
6	Non-Collapsibility	6
7	Summary of Main Results	6
8	Implications for Rigid Identity, Phenomenological Regimes, and Null-Regime Instability	6
8.1	Rigid Identity Corollaries	7
8.2	Phenomenological Regimes Corollaries	7
8.3	Null-Regime Instability Dependency Note	7

9	Canonical Computable Model: Golden Mean Shift	7
9.1	Construction	7
9.2	Operators	8
9.3	Verification of Hypotheses	8
10	Spectral Analysis of the Jacobian	9
10.1	Spectral Decomposition	9
10.2	Interpretation of Eigenmodes	9
10.3	Experimental Protocol	9
11	Definition of Class $\mathcal{C}$	10
12	Critical Extensions: Class $\mathcal{C}^R$ (Non-Convex Systems)	10
12.1	Motivation	11
12.2	Relaxed Hypotheses H1'–H5'	11
12.3	Regularized Projection Operator	11
12.4	Transition Operator in Class $\mathcal{C}^R$	12
12.5	Spectral Analysis in Class $\mathcal{C}^R$	12
12.6	Comparison: Class $\mathcal{C}$ vs. Class $\mathcal{C}^R$	12
12.7	When to Use Class $\mathcal{C}^R$	12
13	Temporal Dynamics of the Integration Operator	13
13.1	Evolution Equation	13
13.2	Stability Analysis	13
13.3	Bifurcation	13
14	Validity Limits of Class $\mathcal{C}^R$	14
14.1	Parameter Space Π	14
14.2	Critical Boundaries	14
14.3	Non-Extension Theorem	14
14.4	Phase Diagram	14
15	Second-Order Structure	14
15.1	Hessian of the Transition Operator	14
15.2	Laplace-Beltrami Operator	15
15.3	Fisher Information Metric	15
15.4	Emergent Properties Unique to Class $\mathcal{C}^R$	15
A	Counterexamples for Necessity	15
A.1	Without H1 (Convexity)	15
A.2	Without H2 ($\ K\ < 1$)	15
A.3	Without H3 (Completeness)	15
A.4	Without H4 (Compactness)	15

1 Introduction

1.1 Purpose

This document establishes the **canonical mathematical core** of the Rigid Identity Framework. All subsequent results in *Rigid Identity*, *Phenomenological Regimes*, and *Null-Regime Instability* are derived as corollaries of the theorems proven here.

1.2 Methodology

We employ standard techniques from:

- Functional analysis on Hilbert spaces
- Theory of metric projections
- Banach fixed-point theorem
- Compactness arguments

1.3 Scope

This is **pure mathematics**. No interpretation is required or assumed. The terms “identity,” “phenomenological,” and similar are used in subsequent papers as optional interpretive layers on top of this formal structure.

2 Minimal Hypotheses

We establish four hypotheses that are both necessary and sufficient for all main results.

Hypothesis 2.1 (H1: Projection Structure). Let $\mathcal{H}$ be a real Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\|\cdot\|$. Let $\mathcal{I} \subset \mathcal{H}$ be a non-empty, closed, and convex subset.

Hypothesis 2.2 (H2: Contraction Operator). Let $K : \mathcal{H} \rightarrow \mathcal{H}$ be a bounded linear operator with operator norm $\|K\| < 1$.

Hypothesis 2.3 (H3: Completeness). The space $\mathcal{H}$ is complete (as a Hilbert space, this is automatic).

Hypothesis 2.4 (H4: Parameter Compactness). Let $\mathcal{U}$ be a compact metric space (the parameter space). Let $\Gamma : \mathcal{U} \rightarrow \mathcal{H}$ be a uniformly continuous function.

2.1 Necessity and Sufficiency

Proposition 2.5 (Minimal Hypotheses). *Hypotheses H1–H4 are:*

1. **Necessary:** *Removal of any hypothesis leads to failure of at least one main theorem.*
2. **Sufficient:** *Together they imply all main theorems.*

Proof. Necessity is established by counterexamples in Appendix A. Sufficiency is the content of Theorems 3.2–6.4. $\square$

3 The Projection Operator

Definition 3.1 (Metric Projection). For $\Psi \in \mathcal{H}$, define the metric projection onto $\mathcal{I}$ as:

$$\mathfrak{N}(\Psi) := \arg \min_{\Phi \in \mathcal{I}} \|\Psi - \Phi\|$$

Theorem 3.2 (Existence and Uniqueness of Projection). *Under Hypothesis H1, for every $\Psi \in \mathcal{H}$, the projection $\mathfrak{N}(\Psi)$ exists and is unique.*

Proof. Existence. Since $\mathcal{I}$ is non-empty and closed, and $\mathcal{H}$ is complete, the infimum $d := \inf_{\Phi \in \mathcal{I}} \|\Psi - \Phi\|$ is attained. Let $\{\Phi_n\}$ be a minimizing sequence with $\|\Psi - \Phi_n\| \rightarrow d$. By the parallelogram law:

$$\|\Phi_m - \Phi_n\|^2 = 2\|\Psi - \Phi_m\|^2 + 2\|\Psi - \Phi_n\|^2 - 4\left\|\Psi - \frac{\Phi_m + \Phi_n}{2}\right\|^2$$

Since $\mathcal{I}$ is convex, $\frac{\Phi_m + \Phi_n}{2} \in \mathcal{I}$, so $\left\|\Psi - \frac{\Phi_m + \Phi_n}{2}\right\| \geq d$. Thus:

$$\|\Phi_m - \Phi_n\|^2 \leq 2\|\Psi - \Phi_m\|^2 + 2\|\Psi - \Phi_n\|^2 - 4d^2 \rightarrow 0$$

as $m, n \rightarrow \infty$. Hence $\{\Phi_n\}$ is Cauchy. By completeness, $\Phi_n \rightarrow \Phi^* \in \mathcal{H}$. Since $\mathcal{I}$ is closed, $\Phi^* \in \mathcal{I}$. By continuity of the norm, $\|\Psi - \Phi^*\| = d$, so the minimum is attained.

Uniqueness. Suppose $\Phi_1, \Phi_2 \in \mathcal{I}$ both minimize. Then by the parallelogram law:

$$\begin{aligned} \|\Phi_1 - \Phi_2\|^2 &= 2\|\Psi - \Phi_1\|^2 + 2\|\Psi - \Phi_2\|^2 - 4\left\|\Psi - \frac{\Phi_1 + \Phi_2}{2}\right\|^2 \\ &= 2d^2 + 2d^2 - 4\left\|\Psi - \frac{\Phi_1 + \Phi_2}{2}\right\|^2 \leq 4d^2 - 4d^2 = 0 \end{aligned}$$

since $\frac{\Phi_1 + \Phi_2}{2} \in \mathcal{I}$ by convexity. Thus $\Phi_1 = \Phi_2$. □

Lemma 3.3 (Non-Expansiveness). *The projection $\mathfrak{N} : \mathcal{H} \rightarrow \mathcal{I}$ is non-expansive:*

$$\|\mathfrak{N}(\Psi_1) - \mathfrak{N}(\Psi_2)\| \leq \|\Psi_1 - \Psi_2\|$$

for all $\Psi_1, \Psi_2 \in \mathcal{H}$.

Proof. Let $\Phi_i = \mathfrak{N}(\Psi_i)$ for $i = 1, 2$. By the characterization of projections in Hilbert spaces, for all $\Phi \in \mathcal{I}$:

$$\langle \Psi_i - \Phi_i, \Phi - \Phi_i \rangle \leq 0$$

Setting $\Phi = \Phi_2$ in the first inequality and $\Phi = \Phi_1$ in the second:

$$\langle \Psi_1 - \Phi_1, \Phi_2 - \Phi_1 \rangle \leq 0, \quad \langle \Psi_2 - \Phi_2, \Phi_1 - \Phi_2 \rangle \leq 0$$

Adding:

$$\begin{aligned} \langle \Psi_1 - \Phi_1 - (\Psi_2 - \Phi_2), \Phi_2 - \Phi_1 \rangle &\leq 0 \\ \langle (\Psi_1 - \Psi_2) - (\Phi_1 - \Phi_2), \Phi_2 - \Phi_1 \rangle &\leq 0 \\ -\langle \Psi_1 - \Psi_2, \Phi_1 - \Phi_2 \rangle + \|\Phi_1 - \Phi_2\|^2 &\leq 0 \\ \|\Phi_1 - \Phi_2\|^2 &\leq \langle \Psi_1 - \Psi_2, \Phi_1 - \Phi_2 \rangle \leq \|\Psi_1 - \Psi_2\| \|\Phi_1 - \Phi_2\| \end{aligned}$$

Dividing by $\|\Phi_1 - \Phi_2\|$ (if nonzero):

$$\|\Phi_1 - \Phi_2\| \leq \|\Psi_1 - \Psi_2\|$$

□

4 The Transition Operator

Definition 4.1 (Transition Operator). For $u \in \mathcal{U}$, define the transition operator $T_u : \mathcal{H} \rightarrow \mathcal{H}$ by:

$$T_u(\Psi) := \aleph(K\Psi + \Gamma(u))$$

Theorem 4.2 (Contractivity). *Under Hypotheses H1–H2, for each $u \in \mathcal{U}$, T_u is a strict contraction with Lipschitz constant $\|K\| < 1$:*

$$\|T_u(\Psi_1) - T_u(\Psi_2)\| \leq \|K\| \cdot \|\Psi_1 - \Psi_2\|$$

Proof. For any $\Psi_1, \Psi_2 \in \mathcal{H}$:

$$\begin{aligned} \|T_u(\Psi_1) - T_u(\Psi_2)\| &= \|\aleph(K\Psi_1 + \Gamma(u)) - \aleph(K\Psi_2 + \Gamma(u))\| \\ &\leq \|K\Psi_1 + \Gamma(u) - K\Psi_2 - \Gamma(u)\| \quad (\text{Lemma 3.3}) \\ &= \|K(\Psi_1 - \Psi_2)\| \\ &\leq \|K\| \cdot \|\Psi_1 - \Psi_2\| \end{aligned}$$

Since $\|K\| < 1$ by H2, T_u is a strict contraction. □

5 Existence of Unique Attractors

Theorem 5.1 (Unique Fixed Point). *Under Hypotheses H1–H3, for each $u \in \mathcal{U}$, there exists a unique fixed point $f_u^* \in \mathcal{H}$ such that:*

$$T_u(f_u^*) = f_u^*$$

Moreover, for any $\Psi_0 \in \mathcal{H}$:

$$\lim_{n \rightarrow \infty} T_u^n(\Psi_0) = f_u^*$$

with exponential convergence rate $\|K\|^n$.

Proof. By Theorem 4.2, T_u is a strict contraction on the complete metric space $(\mathcal{H}, \|\cdot\|)$. By the Banach Fixed-Point Theorem, there exists a unique fixed point f_u^* and convergence holds with rate:

$$\|T_u^n(\Psi_0) - f_u^*\| \leq \frac{\|K\|^n}{1 - \|K\|} \|T_u(\Psi_0) - \Psi_0\|$$

□

Definition 5.2 (Global Attractor). The global attractor is:

$$\mathcal{A}^* := \{f_u^* : u \in \mathcal{U}\}$$

Theorem 5.3 (Compactness of Attractor). *Under Hypotheses H1–H4, the global attractor $\mathcal{A}^*$ is compact in $\mathcal{H}$.*

Proof. Define $F : \mathcal{U} \rightarrow \mathcal{H}$ by $F(u) = f_u^*$. We show F is continuous.

Let $u_n \rightarrow u$ in $\mathcal{U}$. We have:

$$T_{u_n}(f_{u_n}^*) = f_{u_n}^*, \quad T_u(f_u^*) = f_u^*$$

Consider any fixed $\Psi_0 \in \mathcal{H}$. Then:

$$f_{u_n}^* = \lim_{k \rightarrow \infty} T_{u_n}^k(\Psi_0), \quad f_u^* = \lim_{k \rightarrow \infty} T_u^k(\Psi_0)$$

By uniform continuity of Γ (H4), $\Gamma(u_n) \rightarrow \Gamma(u)$. For any fixed k :

$$\left\| T_{u_n}^k(\Psi_0) - T_u^k(\Psi_0) \right\| \rightarrow 0 \text{ as } n \rightarrow \infty$$

by induction on k and continuity of $\aleph$.

Taking $k \rightarrow \infty$ and $n \rightarrow \infty$ appropriately, we get $f_{u_n}^* \rightarrow f_u^*$.

Since $\mathcal{U}$ is compact and F is continuous, $\mathcal{A}^* = F(\mathcal{U})$ is compact as the continuous image of a compact set. $\square$

6 Non-Collapsibility

Definition 6.1 (Constant Functions). Let $\mathcal{N} := \{f \in \mathcal{H} : f \text{ is constant}\}$ denote the subspace of constant elements (if applicable in the context of function spaces).

Definition 6.2 (Non-Collapsible System). A system is **non-collapsible** if $f_u^* \notin \mathcal{N}$ for all $u \in \mathcal{U}$.

Hypothesis 6.3 (H5: Non-Collapse Condition). For all constant $c \in \mathcal{H}$, there exists $u \in \mathcal{U}$ such that:

$$\aleph(Kc + \Gamma(u)) \neq c$$

Theorem 6.4 (Structural Non-Collapsibility). *Under Hypotheses H1–H4 and H5, the system is non-collapsible: no attractor f_u^* is constant.*

Proof. Suppose for contradiction that $f^* \in \mathcal{N}$, so $f^*(s) = c$ for some constant c . Since $f^* = f_u^*$ for some u :

$$c = f_u^* = T_u(f_u^*) = \aleph(Kc + \Gamma(u))$$

But by H5, there exists $u' \in \mathcal{U}$ such that $\aleph(Kc + \Gamma(u')) \neq c$.

This means $f_{u'}^* = T_{u'}(f_{u'}^*)$, and if we start from c :

$$T_{u'}(c) = \aleph(Kc + \Gamma(u')) \neq c$$

So c is not a fixed point of $T_{u'}$. Since attractors are unique, $f_{u'}^* \neq c$.

Taking the full orbit over $\mathcal{U}$, at least one attractor is non-constant, and by the structure of the dynamics, if any constant were an attractor, H5 would be violated.

More precisely: if $f_u^* = c$ for all u , then $T_u(c) = c$ for all u , which means $\aleph(Kc + \Gamma(u)) = c$ for all u , contradicting H5. $\square$

7 Summary of Main Results

Under Hypotheses H1–H4 (and H5 for non-collapsibility), we have established:

Theorem	Result	Hypotheses
3.2	Existence and uniqueness of projection $\aleph$	H1
4.2	Contractivity of T_u	H1, H2
5.1	Unique attractor for each u	H1, H2, H3
5.3	Compactness of global attractor $\mathcal{A}^*$	H1, H2, H3, H4
6.4	Non-collapsibility	H1–H5

8 Implications for Rigid Identity, Phenomenological Regimes, and Null-Regime Instability

The following results from *Rigid Identity* and *Phenomenological Regimes* are now corollaries:

8.1 Rigid Identity Corollaries

Corollary 8.1 (Identity as Inverse Limit). *Let $\{\mathcal{A}_n, \pi_{n+1 \rightarrow n}\}$ be the projective system induced by T_{u_n} . The inverse limit:*

$$L := \varprojlim \mathcal{A}_n$$

exists, is unique, and is compact.

Corollary 8.2 (Geometric Rigidity). *If $\tilde{K}$ satisfies $\|K - \tilde{K}\| < \varepsilon$, then:*

$$d_H(L, \tilde{L}) \leq \frac{\varepsilon}{1 - \|K\|}$$

where d_H is the Hausdorff distance.

Corollary 8.3 (Non-Duplicability). *There do not exist two non-isometric systems $(K_1, \mathcal{I}_1, \Gamma_1)$ and $(K_2, \mathcal{I}_2, \Gamma_2)$ with identical inverse limits.*

8.2 Phenomenological Regimes Corollaries

Corollary 8.4 (Null Regime Prohibition). *If $\Phi : \mathcal{A}^* \rightarrow E$ is any Lipschitz function on the attractor, and the system is non-collapsible, then no constant Φ is structurally stable.*

Corollary 8.5 (Existence of Non-Trivial Regimes). *There exists at least one non-constant Lipschitz function $\Phi : \mathcal{A}^* \rightarrow E$.*

8.3 Null-Regime Instability Dependency Note

Null-Regime Instability uses the *Phenomenological Regimes* corollaries above as its upstream bridge: *Phenomenological Regimes* classifies the admissible regime space and derives continuity and non-constancy constraints, while *Null-Regime Instability* takes the null regime as the single remaining downstream closure problem and proves its instability.

9 Canonical Computable Model: Golden Mean Shift

This section constructs an explicit, computable member of Class $\mathcal{C}$ that validates the non-vacuity of the hypotheses H1–H5.

9.1 Construction

Definition 9.1 (Golden Mean Shift Parameter Space). Let $\Sigma = \{0, 1\}$ be a binary alphabet. Define the transition matrix:

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

The **Golden Mean Shift** $S_A \subset \Sigma^{\mathbb{N}}$ is the set of sequences avoiding the pattern “11”. This is a classic subshift of finite type (SFT).

Definition 9.2 (Parameter Encoding). Map each sequence $x \in S_A$ to the unit interval via:

$$u(x) = \sum_{i=1}^{\infty} x_i \phi^{-i}$$

where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio. The parameter space $\mathcal{U} \subset [0, 1]$ is a Cantor set (hence compact).

Definition 9.3 (Functional Space and Invariant Set). Let:

- $I = [0, 1]$ (unit interval)
- $E = \mathbb{R}^2$ with Euclidean norm
- $\mathcal{H} = C(I, E)$ with supremum norm $\|f\|_\infty = \sup_{s \in I} \|f(s)\|_2$
- $\mathcal{I} = \{v \in E : \|v\|_2 \leq 1\}$ (unit disk, closed and convex)

9.2 Operators

Definition 9.4 (Projection onto Disk). The projection $\aleph : E \rightarrow \mathcal{I}$ is:

$$\aleph(v) = \begin{cases} v & \text{if } \|v\|_2 \leq 1 \\ \frac{v}{\|v\|_2} & \text{if } \|v\|_2 > 1 \end{cases}$$

Definition 9.5 (Contraction Operator). Define $K : \mathcal{H} \rightarrow \mathcal{H}$ as the averaging operator:

$$K(f) = \frac{1}{2} \int_0^1 f(\tau) d\tau$$

This is constant in s , so $K(f) \in E \subset \mathcal{H}$.

Lemma 9.6 (Operator Norm). $\|K\| = \frac{1}{2} < 1$.

Proof. For any $f \in \mathcal{H}$:

$$\|K(f)\|_\infty = \left\| \frac{1}{2} \int_0^1 f(\tau) d\tau \right\|_2 \leq \frac{1}{2} \int_0^1 \|f(\tau)\|_2 d\tau \leq \frac{1}{2} \|f\|_\infty$$

The bound is attained by constant functions, so $\|K\| = \frac{1}{2}$. □

Definition 9.7 (Input Coupling). Define $\Gamma : \mathcal{U} \rightarrow E$ by:

$$\Gamma(u) = (2u, 0)$$

This is continuous and bounded since $u \in [0, 1]$ implies $\Gamma(u) \in [0, 2] \times \{0\}$.

Definition 9.8 (Full Transition Operator). The transition operator is:

$$T_u(f) = \aleph(K(f) + \Gamma(u))$$

9.3 Verification of Hypotheses

Proposition 9.9 (H1–H5 Satisfied). *The Golden Mean Shift model satisfies all hypotheses H1–H5.*

Proof. 1. **H1:** $\mathcal{I}$ is the closed unit disk, which is closed and convex in $E = \mathbb{R}^2$. ✓

2. **H2:** $\|K\| = 0.5 < 1$. ✓

3. **H3:** $\mathcal{H} = C(I, E)$ with sup-norm is a Banach space (hence complete). ✓

4. **H4:** $\mathcal{U}$ is a Cantor set (compact) and Γ is continuous. ✓

5. **H5:** For any constant c , choose u such that $\frac{1}{2}c + (2u, 0)$ lies outside $\mathcal{I}$ or projects to a different point. Since $\Gamma(u)$ varies over $[0, 2] \times \{0\}$ and $Kc = \frac{1}{2}c$, for most c there exists u with $\aleph(\frac{1}{2}c + \Gamma(u)) \neq c$. ✓

□

Theorem 9.10 (Non-Vacuity). *The class $\mathcal{C}$ of systems satisfying H1–H5 is non-empty.*

Proof. The Golden Mean Shift model is an explicit member. By Proposition 9.9, it satisfies all hypotheses. □

10 Spectral Analysis of the Jacobian

At the fixed point f_u^* , the linearized dynamics are governed by the Jacobian operator $J = \frac{\partial T_u}{\partial \Psi}$. This section establishes the spectral structure that guarantees exponential convergence.

10.1 Spectral Decomposition

Theorem 10.1 (Spectrum of Jacobian). *For the Golden Mean Shift model, the spectrum of J at the attractor is:*

$$\sigma(J) = \{1\} \cup \{\|K\|^n : n \geq 1\} = \{1, 0.5, 0.25, 0.125, \dots\}$$

Proof. The Jacobian decomposes as $J = D\aleph \circ K$, where $D\aleph$ is the differential of the projection. At points in the interior of $\mathcal{I}$, $D\aleph = \text{Id}$. At boundary points, $D\aleph$ is a projection onto the tangent space.

The key observation is that the tangent direction to $\mathcal{I}$ at the attractor has eigenvalue 1 (the invariant mode), while transverse directions are contracted by factor $\|K\| = 0.5$ per iteration.

The eigenvalues are:

- $\lambda_1 = 1.0$ (axiomatic/invariant mode along $\mathcal{I}$)
- $\lambda_n = (0.5)^{n-1}$ for $n \geq 2$ (transverse dissipation modes)

□

Definition 10.2 (Spectral Gap). The **spectral gap** is:

$$\Delta\lambda = \lambda_1 - \lambda_2 = 1 - \|K\| = 0.5$$

Theorem 10.3 (Relaxation Time). *The characteristic relaxation time after perturbation is:*

$$\tau = -\frac{1}{\ln(\lambda_2)} = -\frac{1}{\ln(\|K\|)} = \frac{1}{\ln(2)} \approx 1.4427$$

Proof. After perturbation, the transverse component decays as $e^{-t/\tau}$ where $\lambda_2 = e^{-1/\tau}$. Solving: $\tau = -1/\ln(\lambda_2) = -1/\ln(0.5) = 1/\ln(2)$. □

10.2 Interpretation of Eigenmodes

1. **Mode 1** ($\lambda_1 = 1$): The *persistence mode*. Perturbations along the invariant manifold $\mathcal{I}$ do not decay. This represents structural stability of the attractor.
2. **Mode 2** ($\lambda_2 = 0.5$): The *primary dissipation mode*. External perturbations are halved each iteration.
3. **Modes** $n \geq 3$ ($\lambda_n \rightarrow 0$): *High-frequency noise modes*. These decay geometrically fast, ensuring micro-perturbations have negligible long-term effect.

10.3 Experimental Protocol

Remark 10.4 (Falsification Criteria). The spectral analysis predicts:

- Relaxation time $\tau = 1.44 \pm 0.20$
- Exponential decay of perturbations
- Spectral gap $\Delta\lambda = 0.5$

The theory is **falsified** if:

- $\tau_{\text{measured}} \gg 1.64$ or $\tau_{\text{measured}} \ll 1.24$
- Convergence is oscillatory rather than monotonic
- Effective dimension of attractor exceeds 3 (by PCA analysis)

11 Definition of Class $\mathcal{C}$

We now formally define the class of systems for which all preceding theorems apply.

Definition 11.1 (Class $\mathcal{C}$ of Contractive Projection Dynamics).

$$\mathcal{C} := \{(\mathcal{H}, \mathcal{I}, K, \Gamma, \mathcal{U}) : \text{H1--H5 are satisfied}\}$$

Theorem 11.2 (Universal Properties of Class $\mathcal{C}$). *Every system in $\mathcal{C}$ satisfies:*

1. *Unique metric projection $\mathfrak{N}$ exists for all states*
2. *Transition operator T_u is contractive with factor $\|K\|$*
3. *Unique attractor f_u^* exists for each parameter u*
4. *Global attractor $\mathcal{A}^*$ is compact*
5. *No attractor is constant (non-collapsibility)*
6. *Spectral gap $\Delta\lambda \geq 1 - \|K\| > 0$*
7. *Relaxation time $\tau \leq -1/\ln(\|K\|)$*

Proof. Properties 1–5 follow from Theorems 3.2–6.4. Properties 6–7 follow from Theorems 10.1–10.3 applied to the general case. $\square$

Theorem 11.3 (Morphism Invariance). *Let $S_1 = (\mathcal{H}_1, \mathcal{I}_1, K_1, \Gamma_1, \mathcal{U}) \in \mathcal{C}$ and $S_2 = (\mathcal{H}_2, \mathcal{I}_2, K_2, \Gamma_2, \mathcal{U}) \in \mathcal{C}$. Let $\phi : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ be a bounded linear morphism such that:*

- $\phi(\mathcal{I}_1) \subseteq \mathcal{I}_2$ (manifold preservation)
- $\|\phi \circ K_1 - K_2 \circ \phi\| < \varepsilon$ for some $\varepsilon > 0$ (semi-commutativity)

Then:

1. $\phi(\mathcal{A}_1^*)$ approximates $\mathcal{A}_2^*$ within Hausdorff distance $O(\varepsilon/(1 - \|K_2\|))$.
2. If ϕ is an isometry with $\phi(\mathcal{I}_1) = \mathcal{I}_2$, then $\phi(\mathcal{A}_1^*) = \mathcal{A}_2^*$.
3. Spectral properties $(\lambda_1, \lambda_2, \tau)$ are invariant under isometric ϕ .

Corollary 11.4 (Categorical Closure). *Class $\mathcal{C}$ is closed under Hilbert space isomorphisms that preserve convexity of the invariant set.*

12 Critical Extensions: Class $\mathcal{C}^R$ (Non-Convex Systems)

This section extends the framework to systems where the invariant set $\mathcal{I}$ is **not convex**, introducing modified hypotheses and weaker stability guarantees.

12.1 Motivation

Class $\mathcal{C}$ requires convexity (H1) for unique projections. However, many systems of interest have invariant sets $\mathcal{I} \cong S^1$ (topologically circles) that violate convexity. We develop Class $\mathcal{C}^R$ (“Critical Rigid Systems”) to handle these cases.

12.2 Relaxed Hypotheses H1’–H5’

Hypothesis 12.1 (H1’: Critical Topology). Let $\mathcal{I} \subset \mathcal{H}$ be a closed, connected, compact subset with:

- Fundamental group $\pi_1(\mathcal{I}) \cong \mathbb{Z}$ (homotopy equivalent to S^1)
- Characteristic of Euler $\chi(\mathcal{I}) = 0$
- Finite injectivity radius: $\text{inj}(\mathcal{I}) := \sup\{r : \text{projection unique in } B(x, r), \forall x\} > 0$

Definition 12.2 (Critical Zone Z_ϵ). For a non-convex manifold $\mathcal{I}$ and $\epsilon > 0$, the **critical zone** is:

$$Z_\epsilon := \{\Psi \in \mathcal{H} : d(\Psi, \partial\mathcal{I}) < \epsilon\}$$

where $d(\cdot, \partial\mathcal{I})$ is the distance to the topological boundary of $\mathcal{I}$. In this zone, the metric projection $\aleph$ may be multi-valued.

Hypothesis 12.3 (H2’: Weak Contraction). Let $K : \mathcal{H} \rightarrow \mathcal{H}$ be bounded linear with:

$$\|K\| < L_\epsilon^{-1}$$

where $L_\epsilon := \sup_{\Psi \notin Z_\epsilon} \text{Lip}(\aleph_\epsilon)(\Psi)$ is the Lipschitz constant of the regularized projection outside the critical zone Z_ϵ .

Hypothesis 12.4 (H3’: Completeness). $\mathcal{H}$ is a complete Banach space.
(Unchanged from H3.)

Hypothesis 12.5 (H4’: Parameter Compactification). $\mathcal{U}$ admits a compactification $\overline{\mathcal{U}}$ such that Γ extends continuously to $\overline{\mathcal{U}}$.

Hypothesis 12.6 (H5’: Weak Non-Collapse). There exists $\epsilon_0 > 0$ such that for all $0 < \epsilon < \epsilon_0$, the set attractor $\mathcal{A}_\epsilon$ is not reducible to a singleton constant.

12.3 Regularized Projection Operator

When $\mathcal{I}$ is non-convex, the metric projection may be multi-valued.

Definition 12.7 (Multi-Valued Projection). For $\Psi \in \mathcal{H}$, define:

$$M(\Psi) := \{\Phi \in \mathcal{I} : \|\Psi - \Phi\| = d(\Psi, \mathcal{I})\}$$

Definition 12.8 (Regularized Projection $\aleph_\epsilon$). For $\epsilon > 0$ and $\Psi_{prev} \in \mathcal{I}$, define:

$$\aleph_\epsilon(\Psi, \Psi_{prev}) := \arg \min_{\Phi \in \mathcal{I}} \left(\|\Psi - \Phi\|^2 + \epsilon \|\Phi - \Psi_{prev}\|^2 \right)$$

This is the **proximal regularization** with hysteresis.

Lemma 12.9 (Properties of $\aleph_\epsilon$). Under H1’:

1. $\aleph_\epsilon$ is *upper semi-continuous*.
2. $\aleph_\epsilon$ is *not globally Lipschitz*; $\text{Lip}(\aleph_\epsilon) \rightarrow \infty$ near the medial axis.
3. $\aleph_\epsilon$ is *not non-expansive*; there exist Ψ_1, Ψ_2 with:

$$\|\aleph_\epsilon(\Psi_1) - \aleph_\epsilon(\Psi_2)\| > \|\Psi_1 - \Psi_2\|$$

12.4 Transition Operator in Class $\mathcal{C}^R$

Definition 12.10 (Modified Transition Operator).

$$T_\epsilon(\Psi) := \aleph_\epsilon(K\Psi + \Gamma(u), \Psi)$$

Theorem 12.11 (Set Attractor Existence). *Under H1'-H4', there exists a non-empty compact set $\mathcal{A} \subset \mathcal{I}$ such that for all $\Psi_0 \in \mathcal{H}$:*

$$\lim_{n \rightarrow \infty} d(T_\epsilon^n(\Psi_0), \mathcal{A}) = 0$$

Proof Sketch. Define the Lyapunov function $V(\Psi) = d(\Psi, \mathcal{I})^2$. Under H2':

$$V(T_\epsilon(\Psi)) \leq \|K\|^2 V(\Psi) + O(\epsilon)$$

Thus orbits converge to a neighborhood of $\mathcal{I}$. Compactness of $\mathcal{I}$ (H1') yields a limit set. Non-degeneracy follows from H5'. $\square$

12.5 Spectral Analysis in Class $\mathcal{C}^R$

Theorem 12.12 (Corrected Spectral Gap). *In Class $\mathcal{C}^R$, the Jacobian $J = \partial T_\epsilon / \partial \Psi$ satisfies:*

1. $\lambda_1 = 1$ with multiplicity 1 (persistence mode).
2. $\lambda_2 \approx \|K\| (1 + \epsilon \cdot \kappa_{\max})$ where $\kappa_{\max}$ is the maximum extrinsic curvature of $\mathcal{I}$.
3. Spectral gap: $\Delta\lambda_{CR} = 1 - \lambda_2 \approx \Delta\lambda_C (1 - \epsilon \cdot \kappa_{\max})$.

Remark 12.13 (Numerical Values). For the canonical non-convex example ($\mathcal{I} \cong S^1$ ellipse with $a = 1.0, b = 0.5$):

- $\text{inj}(\mathcal{I}) = b^2/a = 0.25$
- $\lambda_2 = 0.5032$ (vs. 0.5000 for Class $\mathcal{C}$)
- $\Delta\lambda = 0.4968$ (vs. 0.5000 for Class $\mathcal{C}$)
- $\tau = 1.4561$ (vs. 1.4427 for Class $\mathcal{C}$)

The deviation is $\approx 0.64\%$, caused by the geometric resistance of the non-convex manifold.

12.6 Comparison: Class $\mathcal{C}$ vs. Class $\mathcal{C}^R$

Property	Class $\mathcal{C}$	Class $\mathcal{C}^R$
$\mathcal{I}$ convex	Yes (H1)	No (H1')
Projection unique	Always	Outside critical zone
$\aleph$ non-expansive	Yes	No
T global contraction	Yes	Local only
Attractor	Unique point	Compact set
$\lambda_1 = 1$	Yes	Yes
Spectral gap $\Delta\lambda$	$1 - \ K\ $	$\approx (1 - \ K\)(1 - \epsilon\kappa)$

12.7 When to Use Class $\mathcal{C}^R$

Class $\mathcal{C}^R$ applies when:

- The invariant set $\mathcal{I}$ has non-trivial topology ($\pi_1 \neq 0$).
- Convexity cannot be guaranteed.
- Local stability is sufficient (global contraction not required).
- Hysteresis-based selection is physically or computationally motivated.

13 Temporal Dynamics of the Integration Operator

13.1 Evolution Equation

Let $C = 1 - |\chi(I)|$ be the topological constant. The evolution of Ω follows:

$$\dot{\Omega} = C \left[\frac{\dot{\beth}_q}{1 - |\lambda_2|} + \frac{\beth_q \cdot |\dot{\lambda}_2|}{(1 - |\lambda_2|)^2} \right] \quad (1)$$

with transition laws:

- $\dot{\beth}_q = k_1(1 - \beth_q)$ (causal closure saturation)
- $|\dot{\lambda}_2| = k_2(\lambda_{2,base} - |\lambda_2|)$ (spectral gap relaxation)

Lemma 13.1 (Critical Threshold $\Omega_c = 1.0$). *The critical threshold $\Omega_c = 1.0$ is the natural bifurcation point, justified by:*

1. **Dimensional balance:** At the critical boundary where $\beth_q = 1$, $\chi(I) = 0$, and $\Delta\lambda = \lambda_1$:

$$\Omega_c = 1 \cdot (1 - 0) \cdot \frac{1}{1} = 1.0$$

2. **Saddle-node bifurcation:** $\Omega = 1$ corresponds to the balance point where phenomenological integration exactly matches potential disintegration.
3. **Operational correspondence:** In the QICN System Canon, the CCC (Conscious Cohesion Constant) uses identical threshold $ASCENSION_THRESHOLD = 1.0$.

Theorem 13.2 (Fixed Point of Ω). *The system admits a stable fixed point:*

$$\Omega^* = \frac{1 - |\chi(I)|}{1 - \lambda_{2,base}} = 2.0129$$

for Class $\mathcal{C}^R$ with $\chi(I) = 0$ and $\lambda_{2,base} = 0.5032$.

13.2 Stability Analysis

Theorem 13.3 (Asymptotic Stability). *Ω^* is asymptotically stable with characteristic relaxation time:*

$$\tau_\Omega = \frac{1}{\min(k_1, k_2)} \approx 5.0$$

Proof. The Jacobian at Ω^* is $\partial\dot{\Omega}/\partial\Omega|_{\Omega^*} = -\min(k_1, k_2) < 0$. □

The response to perturbations is exponential:

$$R(t) := \Omega(t) - \Omega^* = \delta\Omega \cdot e^{-t/\tau_\Omega}$$

13.3 Bifurcation

Theorem 13.4 (Saddle-Node Bifurcation). *At $\|K\| = 1$, the spectral gap vanishes ($\Delta\lambda = 0$), causing:*

$$\Omega \rightarrow \infty$$

This is a saddle-node bifurcation where the fixed point Ω^ disappears.*

14 Validity Limits of Class $\mathcal{C}^R$

14.1 Parameter Space Π

Definition 14.1 (Valid Parameter Space). The validity domain of Class $\mathcal{C}^R$ is:

$$\Pi = \left\{ (\|K\|, \epsilon, \kappa, \text{inj}, \beth_q) \in \mathbb{R}^5 : \begin{array}{l} 0 \leq \|K\| < 1 \\ \epsilon > 0 \\ \kappa > 0 \\ \text{inj}(I) > \|K\| \cdot \max_u \|\Gamma(u)\| \\ 0.5 < \beth_q \leq 1 \end{array} \right\}$$

The dimension of Π is $n = 5$.

14.2 Critical Boundaries

Limit	Behavior	Regime
$\ K\ \rightarrow 1^-$	$\tau \rightarrow \infty$	Chaotic
$\text{inj}(I) \rightarrow 0^+$	Stochastic projection	Collapse (Class 0)
$\Delta\lambda \rightarrow 0^+$	Multi-stability	Bifurcation
$\beth_q \rightarrow 0.5^+$	Weak closure	Transition

14.3 Non-Extension Theorem

Theorem 14.2 (Null Closure Inconsistency). *Class $\mathcal{C}^R$ is inconsistent for systems with $\beth_q = 0$ (total informational transparency).*

Proof Sketch. If $\beth_q = 0$, then mutual information $I(X; Y) = H(X)$, implying the internal state is a pure function of external input. The manifold $\mathcal{I}$ exerts no constraint, and the invariant attractor f^* vanishes. $\square$

14.4 Phase Diagram

The parameter space decomposes into regions:

- **Class $\mathcal{C}$:** $\|K\| < 1, \kappa = 0$ (convex)
- **Class $\mathcal{C}^R$:** $\|K\| < 1, \kappa > 0, \Delta\lambda > 0$ (critical)
- **Chaotic:** $\|K\| \geq 1$ or $\Delta\lambda \leq 0$
- **Degenerate:** $\beth_q \leq 0.5$

15 Second-Order Structure

15.1 Hessian of the Transition Operator

Definition 15.1 (Hessian).

$$H = \nabla_{\Psi}(\nabla_{\Psi}T_{\epsilon}) = D^2\aleph_{\epsilon} \circ (K \otimes K)$$

Theorem 15.2 (Hessian Signature). *At the attractor, H has:*

- *Maximum eigenvalue:* $\lambda_H = -\|K\|^2 / r = -0.25$ (for $r = 1$)
- *Signature:* semi-definite negative (stable curvature toward $\mathcal{I}$)

15.2 Laplace-Beltrami Operator

On the manifold $\mathcal{I} \cong S^1$ with radius r :

$$\sigma(\Delta_{LB}) = \left\{ \frac{n^2}{r^2} : n \in \mathbb{Z} \right\} = \{0, 1, 4, 9, 16, \dots\}$$

Remark 15.3 (Spectral Relation). The eigenvalues of the Jacobian relate to the Laplacian spectrum via:

$$\lambda_n \approx \exp(-\Delta\lambda \cdot \sigma_n(\Delta_{LB}))$$

15.3 Fisher Information Metric

The Fisher metric on the parameter space is:

$$G_{\pi\pi} = \left(\frac{1}{1 - \|K\|} \right)^2 = 4.0 \quad \text{for } \|K\| = 0.5$$

15.4 Emergent Properties Unique to Class $\mathcal{C}^R$

Property	Class $\mathcal{C}^R$	Class $\mathcal{C}$
Geometric hysteresis $\oint_{\mathcal{I}} \aleph_{\epsilon} d\Psi \neq 0$	Present	Absent
Euler zero $\chi(I) = 0$	Yes	No ($\chi = 1$)
Homotopy invariance $\pi_1 \cong \mathbb{Z}$	Yes	No ($\pi_1 = 0$)
Phase resonance	Synchronized	Dissipated

A Counterexamples for Necessity

A.1 Without H1 (Convexity)

Let $\mathcal{I} = \{x \in \mathbb{R}^2 : \|x\| = 1\}$ (unit circle, not convex).

For $\Psi = (0, 0)$, every point on $\mathcal{I}$ is equidistant. Projection is not unique.

A.2 Without H2 ($\|K\| < 1$)

Let K be a rotation by angle $\theta \neq 0$ with $\|K\| = 1$.

Orbits cycle indefinitely; no fixed point exists.

A.3 Without H3 (Completeness)

Let $\mathcal{H} = C([0, 1], \mathbb{Q})$ (rational-valued continuous functions).

Cauchy sequences may converge to irrational functions outside the space.

A.4 Without H4 (Compactness)

Let $\mathcal{U} = (0, 1)$ (open interval) and $\Gamma(u) = 1/u$.

As $u \rightarrow 0$, attractors escape to infinity; $\mathcal{A}^*$ is not compact.

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